Jayanth Dasamantharao

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, *EliteUS Software Solutions* — *Aug 2024 - Present | Piscataway, NJ*

- Designed multi-agent AI systems using **AutoGen & Agentic AI** frameworks, orchestrating **LLM-driven** workflows across cloud environments. Integrated large language models (LLMs) with vector databases(ChromaDb, AzureSearch) and **REST APIs**, enabling dynamic AI agent interactions, fallback mechanisms.
- Implemented scalable **Retrieval-Augmented Generation (RAG) pipelines** combining Amazon Titan and Azure OpenAI embeddings with LLaMA models on AWS Bedrock, optimizing for multi-cloud inference.
- Implemented **MLOps** practices for deploying and monitoring **AI pipelines**, including scalable **CI/CD workflows** for model deployment, versioning and vector database updates across AWS and Azure environments.

Data Science Intern, *Harvest Software Solutions, LLC* — *June 2023 - Sept 2023 | Jacksonville, Florida*

- Applied **predictive models** to operational datasets, providing actionable insights that improved **supply chain efficiency** and cost management by 70% boost in predictive accuracy and ad spend efficiency.
- Leveraged big data tools such as **PySpark, Pandas** to process and analyze datasets exceeding 10 million rows, enabling real-time decision-making in supply chain optimization. **FastAPI** was used to **deploy NLP models.**

Application Development Associate, *Accenture Solutions Pvt. Ltd.* — *June 2021 - Aug 2022 | Hyderabad, India*

Database Analyst, Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Designed and optimized **ETL workflows** using **Tableau Prep** to streamline data preparation processes and implemented **automated data pipelines in Python**, enabling integration of large-scale supply chain data for analysis and reporting.
- Performed large-scale data analysis using **SQL** and **AWS Athena, Redshift** on large scale datasets; visualized insights and built interactive dashboards in **Tableau and Power BI**.
- Increased system reliability by 25% through **DDL optimization, seamless data integration, and performance tuning**, ensuring stable **data warehousing** and supporting data-driven strategies in **supply chain management.**

Research Lab, *Andhra University* — *January 2021 - June 2021 | Visakhapatnam, India*

Self Driving Cars Using Image Processing and Deep Learning

- Engineered **Convolutional Neural Networks** (CNNs) for autonomous lane detection and object recognition using python.
- Leveraged **YOLO and Faster R-CNN** architectures to optimize real-time detection and classification accuracy.
- Enhanced model generalization through **image pre-processing** techniques like **Gaussian blurring and color space conversion**, and integrated **multi-sensor data** from **radar, lidar, and cameras** to improve navigational precision.
- Developed a real-time simulation environment using **Unity3D and a Flask backend** for dynamic parameter monitoring and system validation, enabling high-fidelity testing of autonomous driving functionalities.

TECHNICAL SKILLS

Programming:	Python, R, Scala, MATLAB. **Libraries:** Pytorch, Scikit-learn, Tensorflow, SparkML, Tableau, PowerBI.
Machine Learning:	Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis:	Causal Inference, Clustering, Hypothesis testing, Time-series forecasting, Recommendation Engines.
Database & Cloud:	SQL, PostgreSQL, AWS(S3, Glue, RDS), ETL, BigQuery, MongoDB, Langchain, OneDev, Git, MLOps.
Data Engineering:	Databricks, Azure, Kafka, Apache Spark, Data Warehousing, Hadoop, GCP, Tableau Prep, PowerBI.

EDUCATION

Rutgers University, *Masters in Data Science* — *Sept 2022 - May 2024 | New Brunswick, NJ*

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** Portfolio.

Andhra University, *B.Tech in Electrical and Electronics Engineering* — *2017 - 2021 | Visakhapatnam, India*

RESEARCH PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.